Teacher Wu Kuan | See It at a Glance #1 — sin(θ + 90°) = cos θ

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1. The trap question

Why is sin(θ + 90°) = cos θ?

2. The method

• Rotate the unit-circle triangle by 90°.
• The old horizontal leg (cos θ) becomes the new height (sin(θ + 90°)).
• Also cos(θ + 90°) = −sin θ.

3. Parent check method

Have your child rotate a paper triangle 90° and watch which side becomes the height.

Practice

1. sin(30° + 90°) = ?
2. Write sin(θ + 90°) in terms of cos θ.
3. Use 60° + 90° to find sin 150°.
4. Write cos(θ + 90°) in terms of sin θ.
5. Use 45° + 90° to find sin 135°.

Answers

1. √3/2 (= cos 30°)
2. cos θ
3. 1/2
4. −sin θ
5. √2/2

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